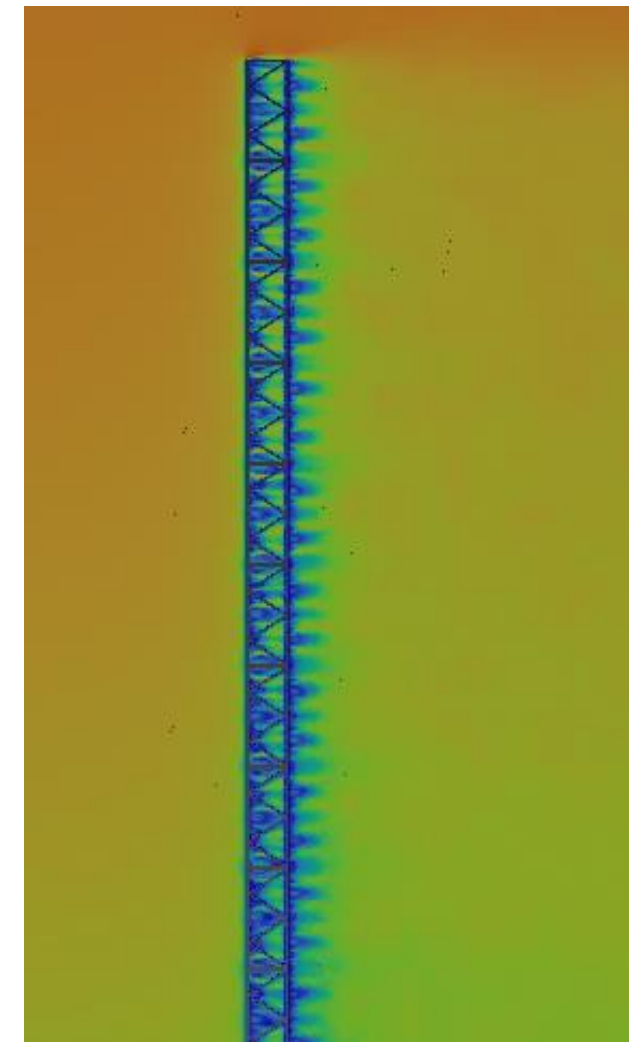
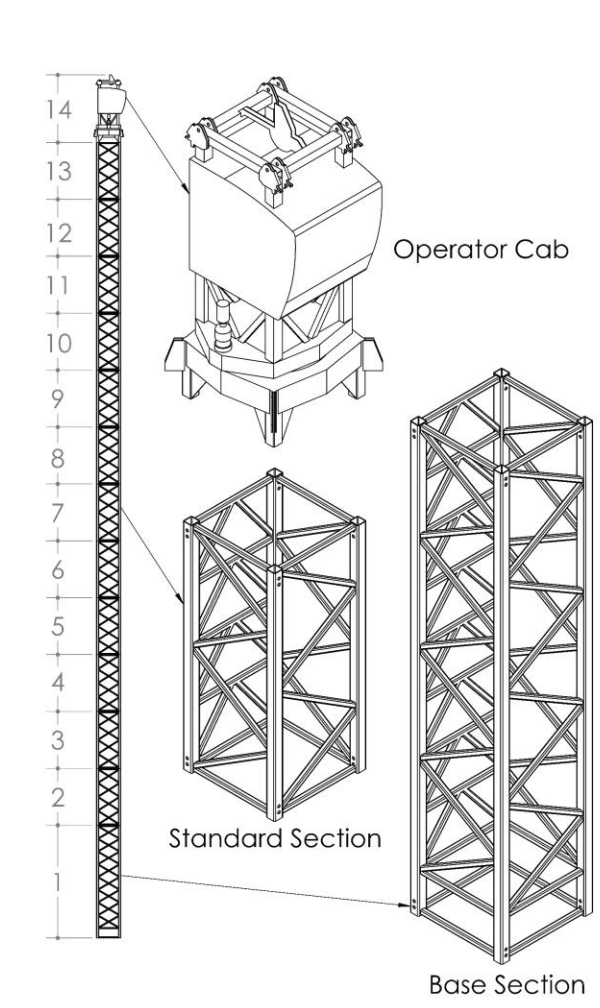


Background & Motivation



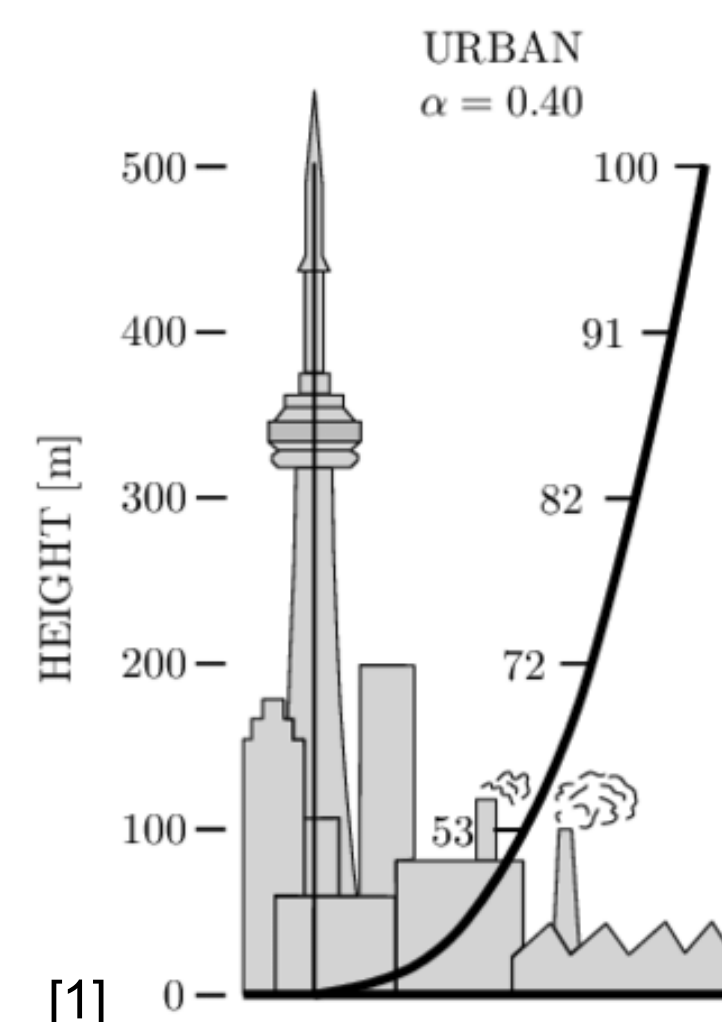
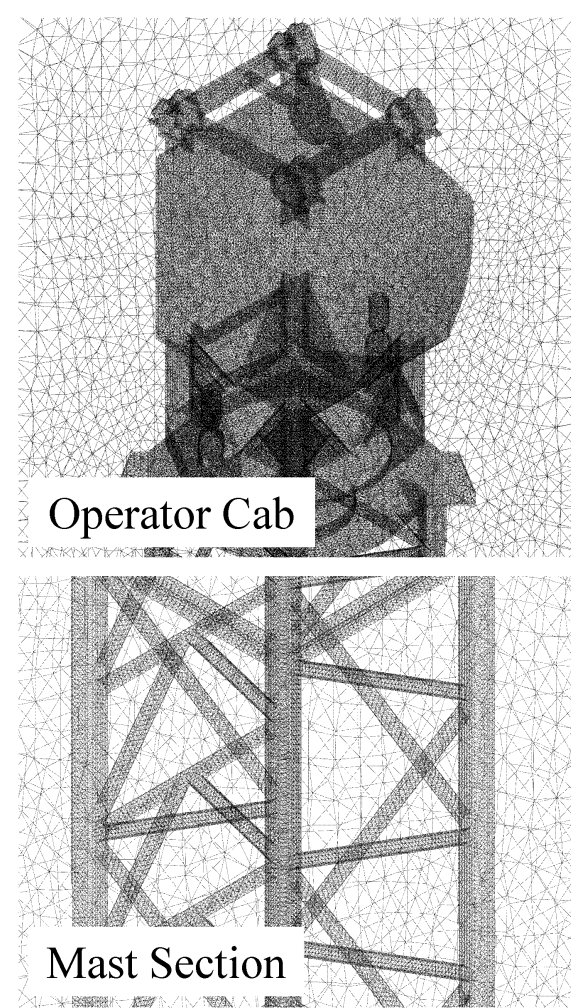
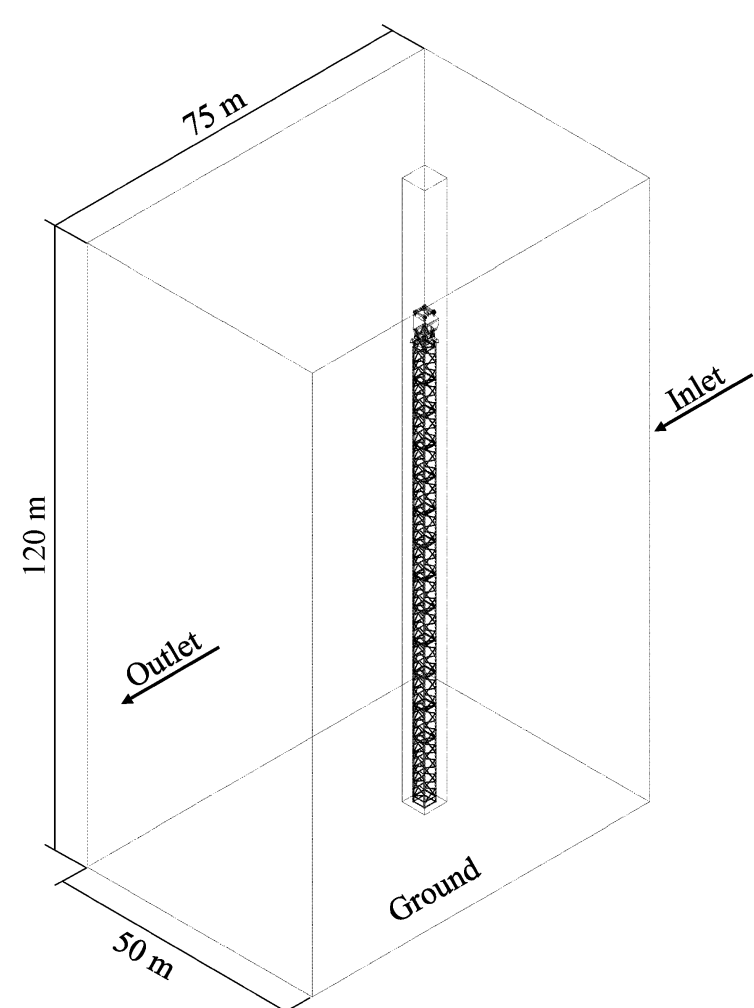
- Wind loading poses a safety risk for tower cranes
- Lattice masts generate complex wake and shielding effects
- CFD captures these physics but is computationally intensive
- Reduced-order models enable rapid, scalable wind-load assessment

Project Objectives

- 1) Use CFD to quantify directional wind loads on tower cranes
- 2) Evaluate the effects of mast height and operator cab
- 3) Develop an area-based decomposition to classify aerodynamic exposure
- 4) Construct a reduced-order model to predict drag and overturning moment from CFD data

Method

Workflow



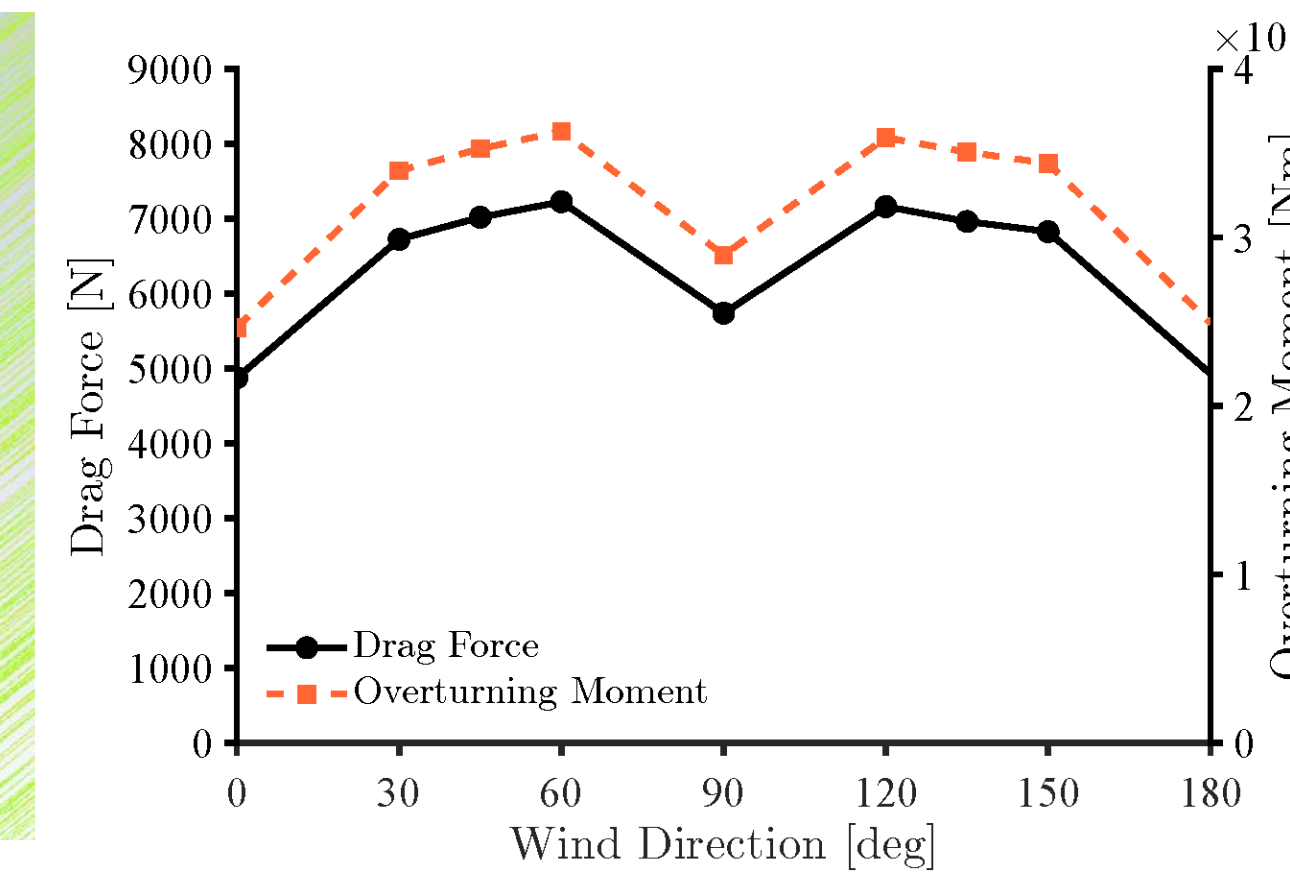
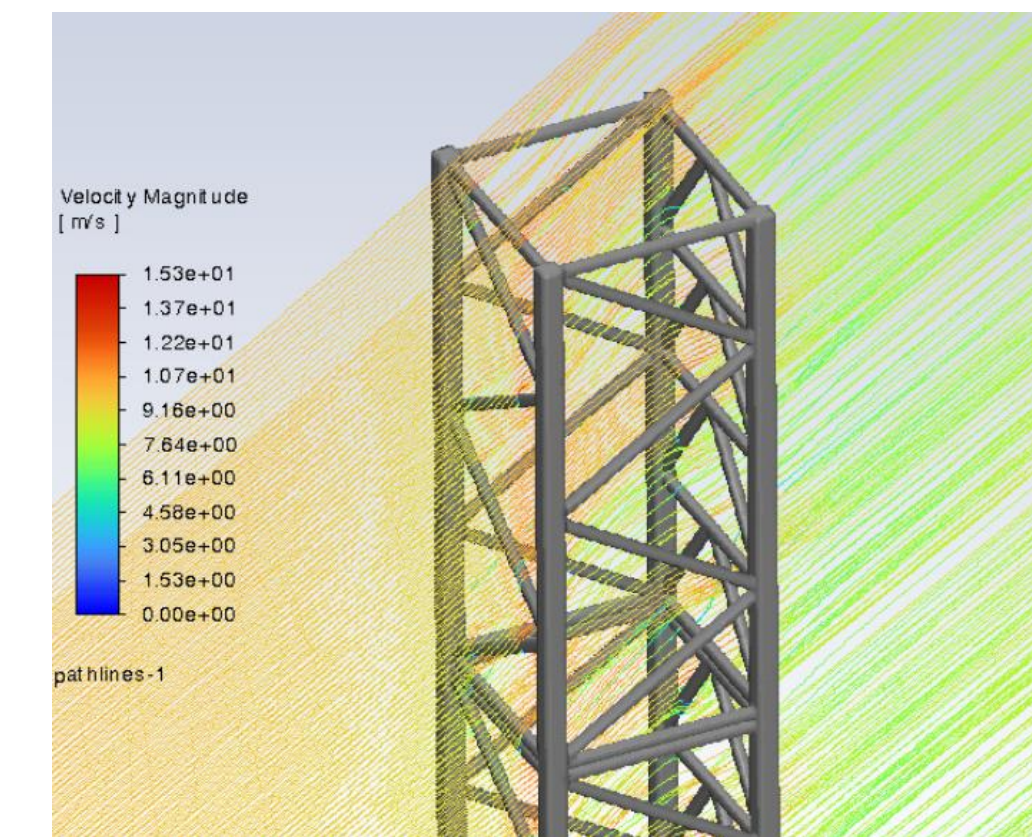
- Parametric geometry with varying configurations and angle
- Steady RANS simulations with Atmospheric Boundary Layer (ABL) inlet: [2]

$$\text{Log-law inlet velocity profile, } U(z) = \frac{u_*}{\kappa} \ln \left(\frac{z + z_0}{z_0} \right)$$

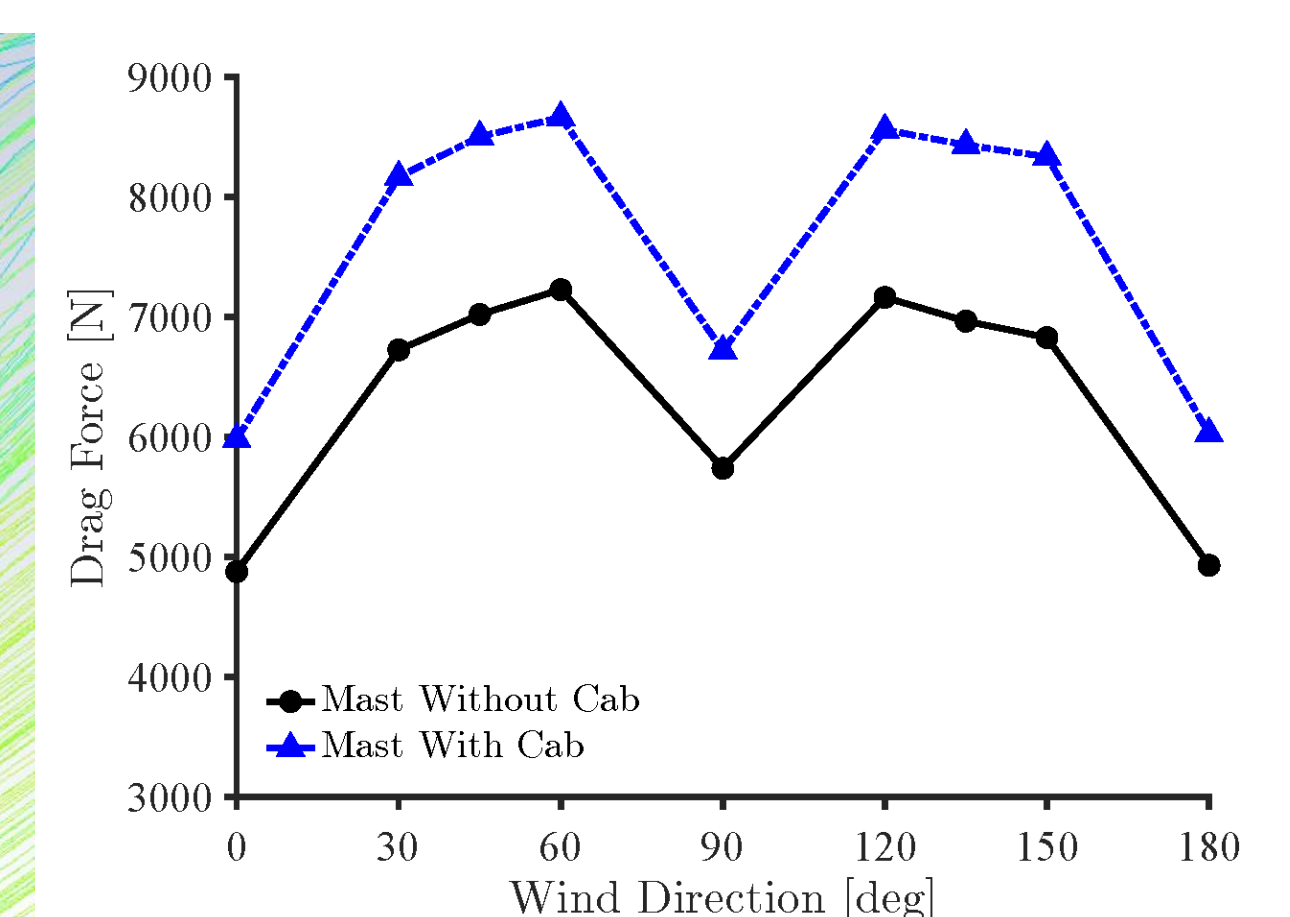
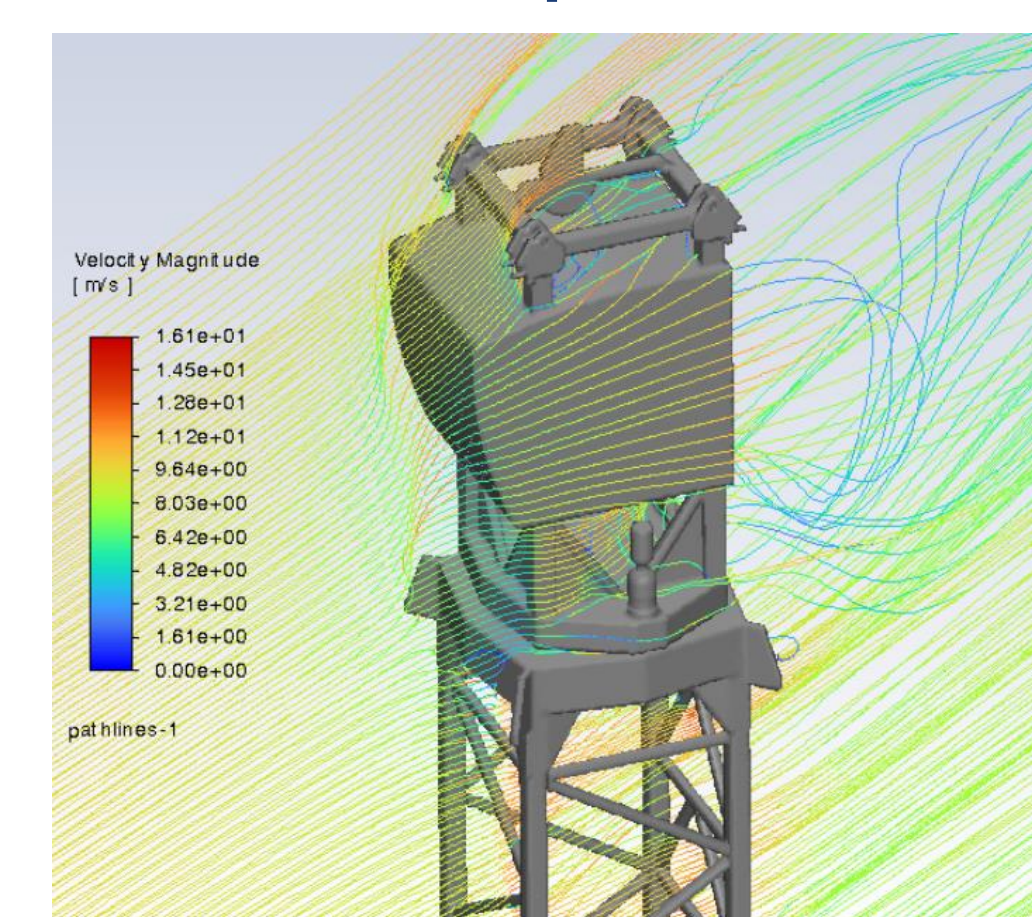
- Evaluate drag forces and overturning moment on crane walls
- Projected area normal to wind direction is decomposed into – First-contact, Second-contact and Shielded areas
- Area based empirical model is built to predict wind loads

Results - CFD

Mast only Configuration

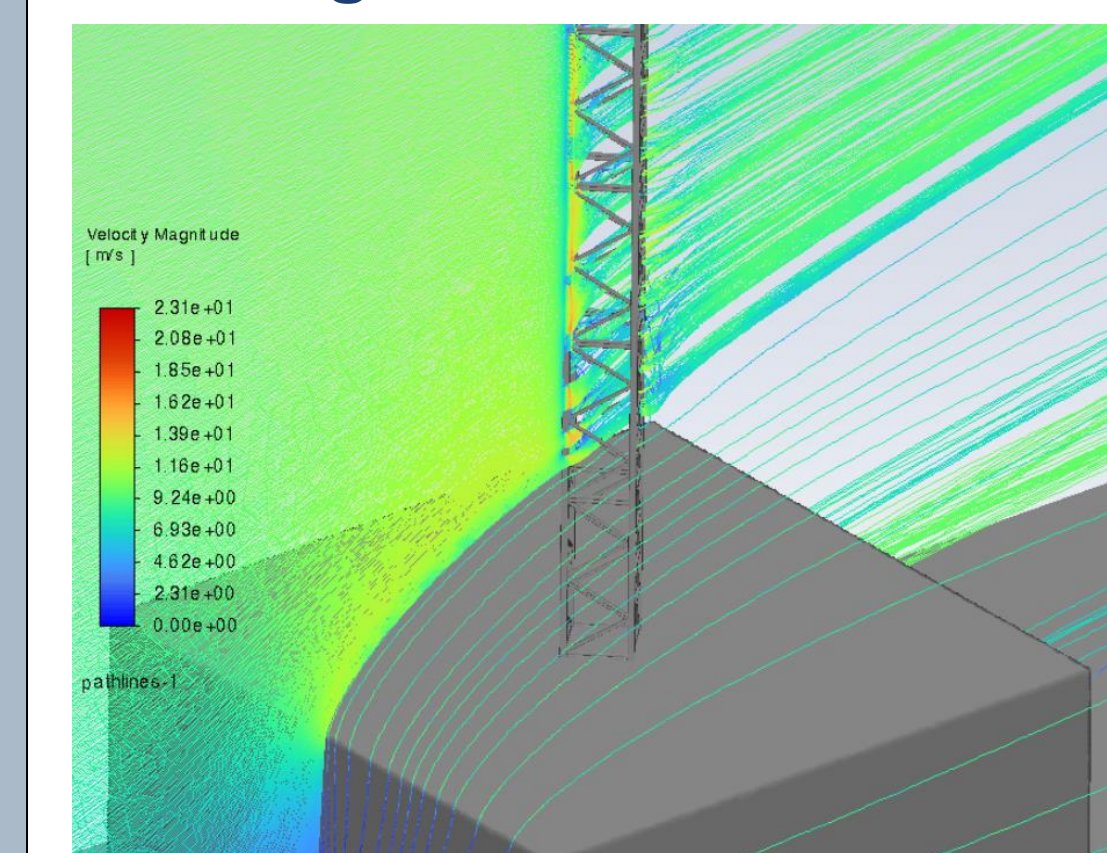


Influence of Operator Cab

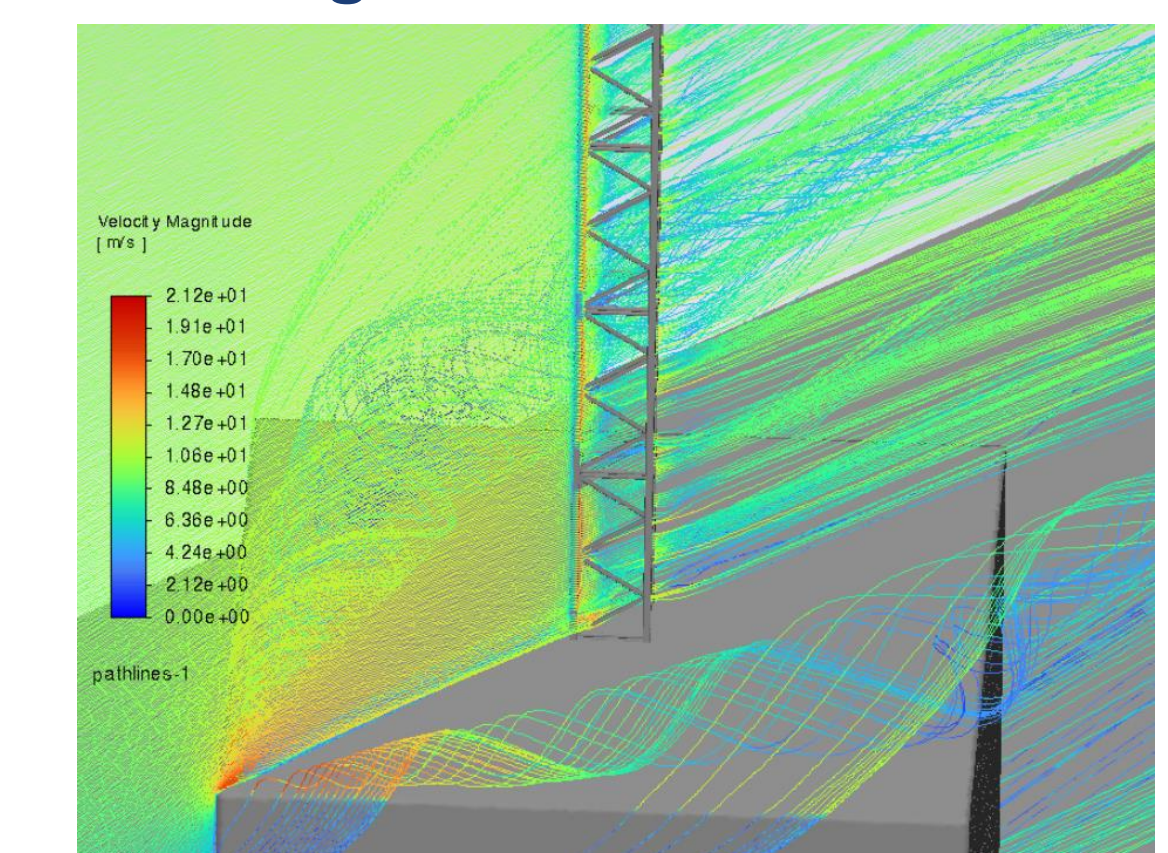


Results – CFD (Building Impact)

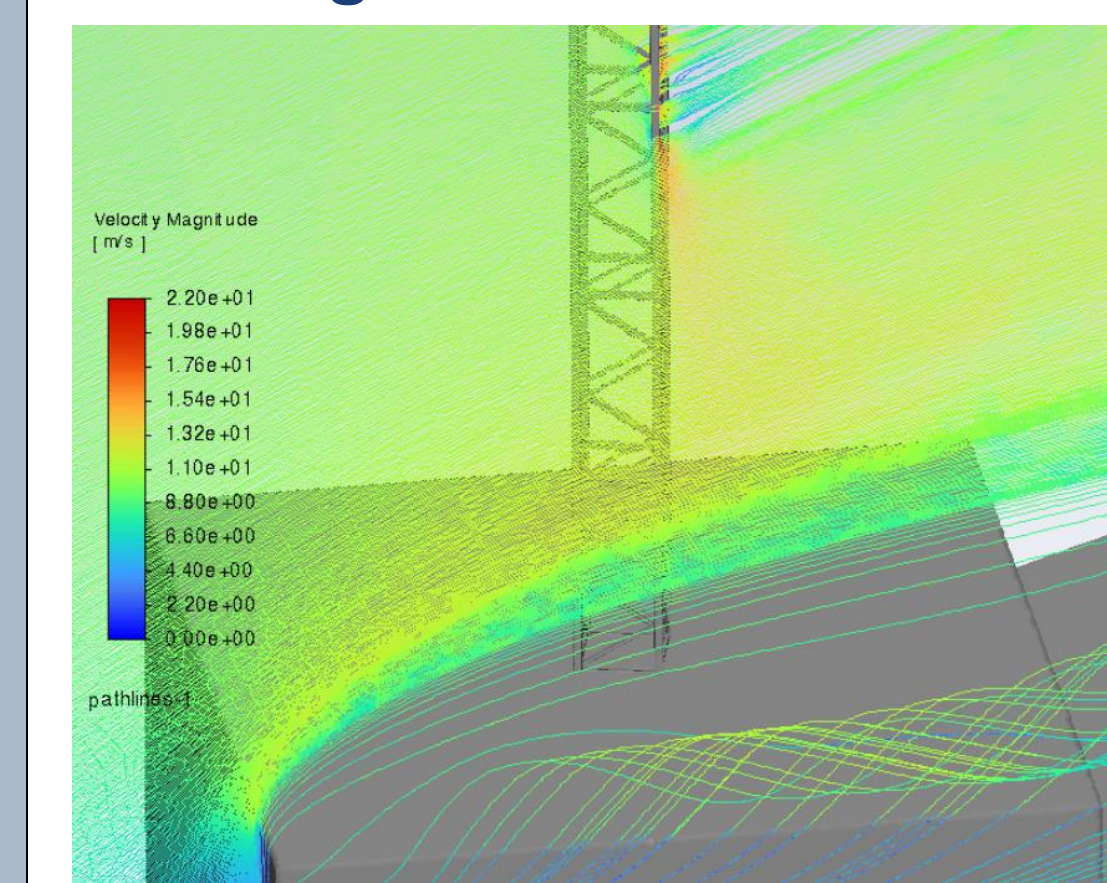
Building 0° & Tower 45°



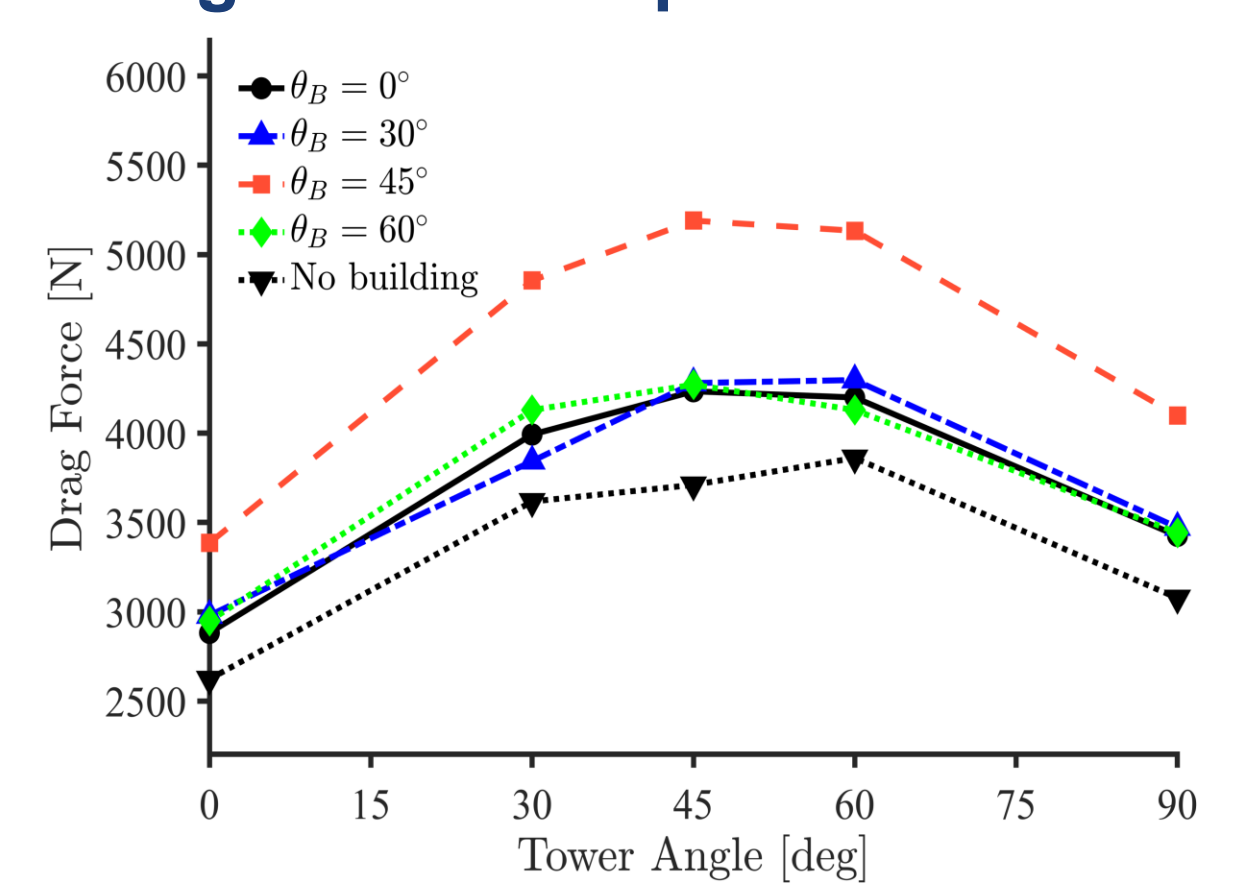
Building 45° & Tower 45°



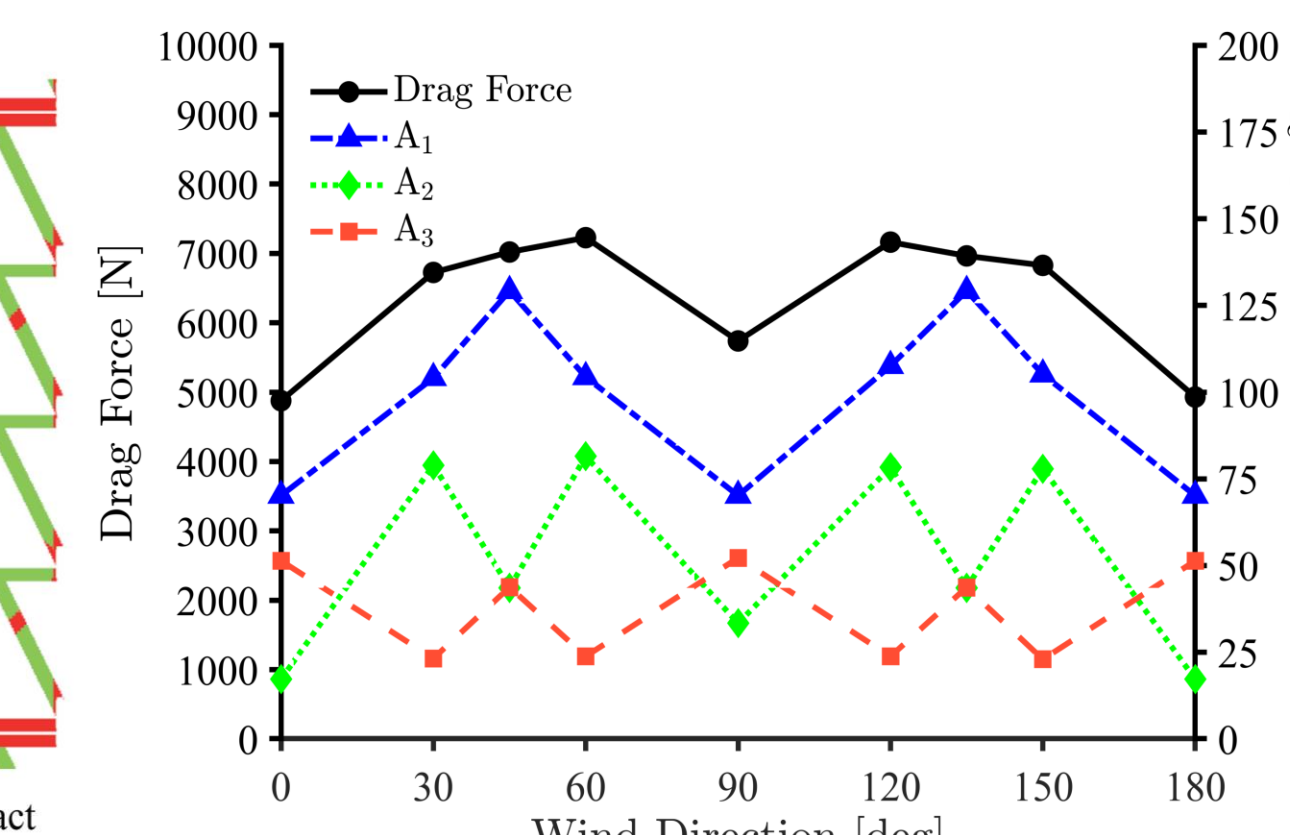
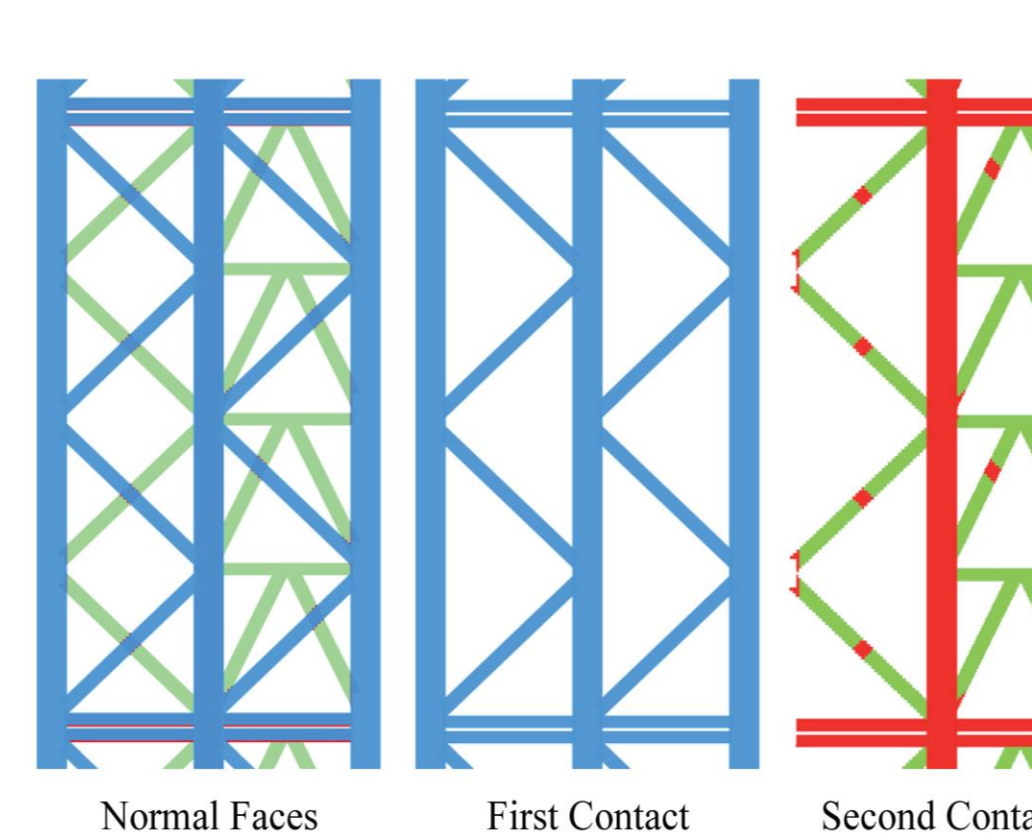
Building 60° & Tower 45°



Drag Force Comparison



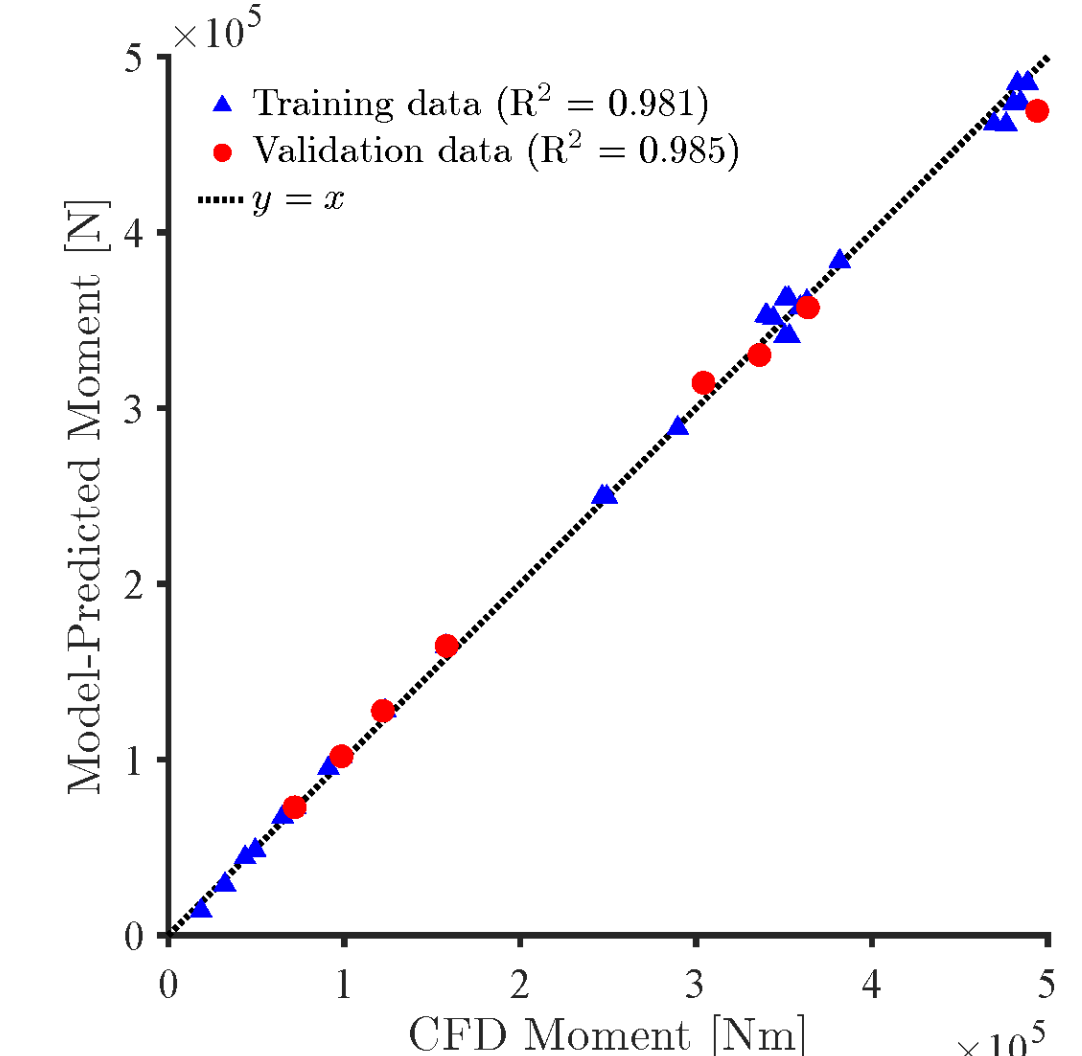
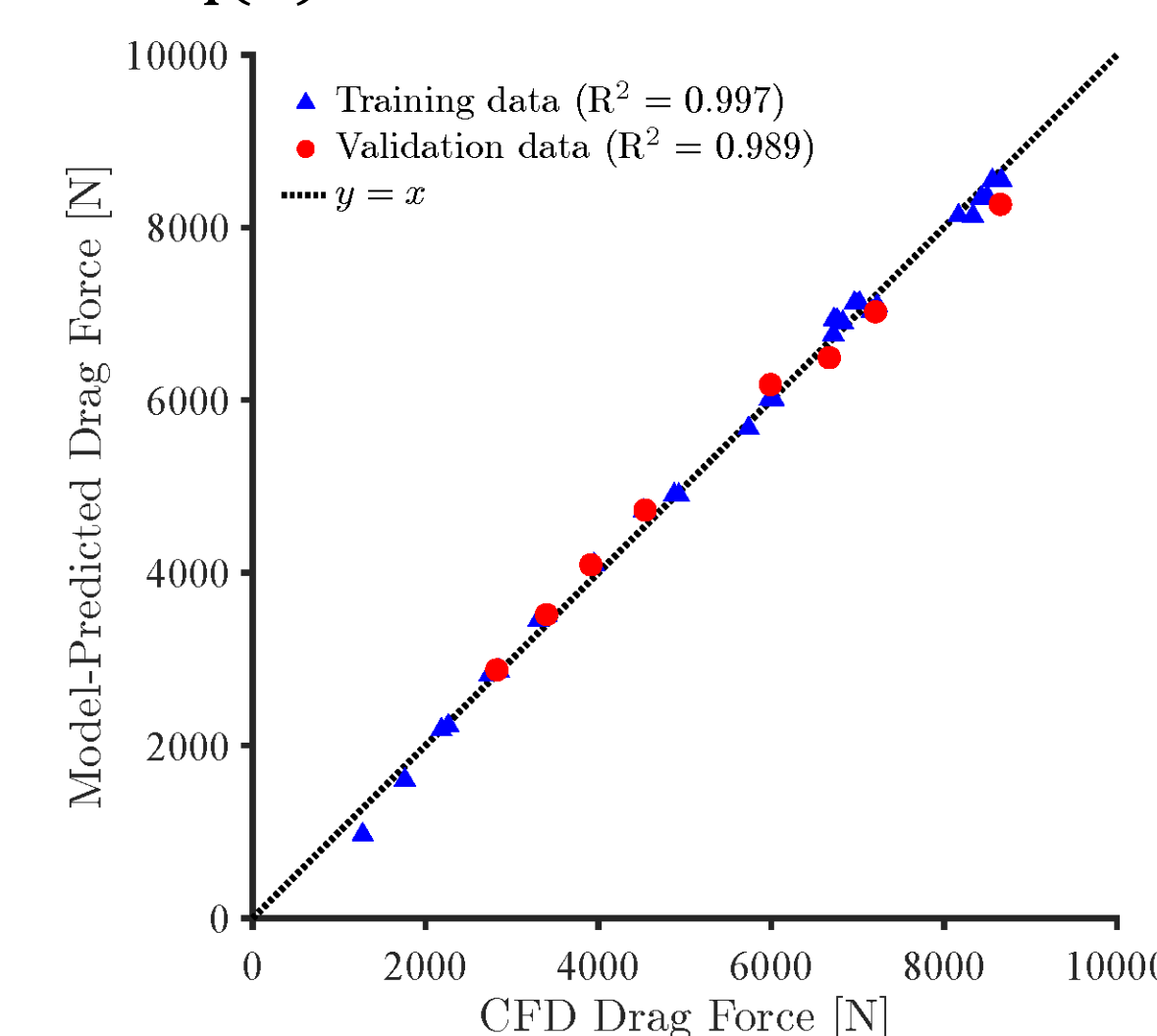
Results – Area Decomposition



Results – Empirical Model

Drag Force Model

$$\frac{F_D(\theta, N_s, I_{cab})}{q(\theta)} = \beta_1 A_1(\theta) + \beta_2 A_2(\theta) + \beta_3 A_3(\theta) + \beta_4 H_{cp} + \beta_5 N_s + \beta_6 I_{cab}$$



Summary / Conclusions

- Oblique wind angles produce peak drag and moments
- Cab contribution exceeds that of single mast section
- Area decomposition captures shielding and exposure effects
- Empirical model achieves high-accuracy rapid prediction

Acknowledgments

Support for the Crane Safety Research Center at Georgia Tech was provided by the Sarah Pantip Wong Family.

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